

GAS PROCESSING & METHANOL COMPLEX PROJECT NO. 11998A

Project	FEED for Gas Gathering Pipelines & Associated Infrastructure
Client	Brass Fertilizer and Petrochemical Company Limited
Originating Company	Epic Atlantic Limited
Document Title	Consumables Requirements & Supply Plan
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Consumables Requirements & Supply Plan

R02	20.09.22	Re-Issued for Review	O. Ige	M. Ekpunobi	A. Duru	
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Revision History

Rev No.	Date	Description of Revision
R01	07.07.22	Issued for Review
R02	20.09.22	Re-issued for Review after incorporating TCE comments
A01		

Revision Philosophy:

- a. All documents for review shall be issued at R01, with subsequent R02, R03, etc as required.
- b. All documents approved for issue or approved for design shall be issued at A01 with subsequent A02, A03, etc. as required. (Management of Change is required for A02, A03, etc.)
for the subsequent project phases, post FEED, it is expected that:
- c. All Detailed Design documents for review shall be issued at D01, with subsequent D02, D03, etc. as required.
- d. All documents approved for construction shall be issued at C01 with subsequent C02, C03, etc. as required. (Management of Change is required for C02, C03, etc.)
- e. All approved "As-Built" documents shall be issued at Z01, with subsequent Z02, Z03, etc as required. (Use versions Z01.1, Z01.2, Z01.3, etc to review the "As-Built" document to Z02).

Restricted

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SUMMARY PAGE

Client:	Brass Fertilizer & Petrochemical Company Limited	
Project:	Gas Processing & Methanol Complex	Project No. 11998A
Facility:	Gas Gathering Pipelines & Associated Infrastructure	Doc No. EA-BFPCL-GPMC-GGPL-FEED- HSE-PLN-002

The administration of this document is in respect of FEED for Gas Gathering Pipelines & Associated Infrastructure for the Gas Processing & Methanol Plant Project of BFPCL.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2. The focus of the present FEED is on Phase 1, while Phase 2 will be implemented later.

It is also a precursor and accompaniment to the FEED study & report that will lead to the project's FID and implementation.

The report has relied on and used data and reference documents provided by the client, and if further documents and information become available, Epic Atlantic reserves the right to update the opinion set out in this report.

For better appreciation, this document should be read in conjunction with the references listed below, which are the Concept Select Study reports that constitute basis for the Conceptual Study.

KEYWORDS: BASIS OF DESIGN, PIPELINE, MANIFOLD, PLATFORM, MULTIPHASE FLOW, ROW, PTS

REFERENCES

Document No.	Description
EA-BFPCL-GPMC-GGPL-FEED-PSE-RPT-001	Upstream Gas Gathering Concept Select & Revalidation Report
EA-BFPCL/BFPP GAS PL/FEED-PRO-RPT-001	Preliminary Simulation – Based on Desktop Route Survey
EA/SPDC-BFPC UGS-GEP/CSM-RPT-002	Gas Gathering & Evacuation Pipelines Concept Screening
EA/SPDC-BFPC UGS-GEP/CSM-RPT-001	Conceptual Study Memorandum
EA-BFPCL-GPMC-GGPL-FEED-GEN-RPT-002	Basis of Design
	FISAS Draft ESIA Report (November 2020)

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Abbreviations

Abbreviation	Meaning
AG	Associated Gas
ANSI	American National Standards Institute
API	American Petroleum Institute
BBOU	Bubouwe Bou Manifold
BFPC	Brass Fertilizer & Petrochemical Company Limited
BLV	Blowdown Valve
CAO	Computer Aided Operations
ESD	Emergency Shut Down
FEED	Front End Engineering Design
FLB	Field Logistics Base
FLM	First Line Maintenance
FQI	Flow Quantity Indicator
GSPA	Gas Supply & Purchase Agreement
HP	High Pressure
HSSE	Health Safety Security and Environment
LCP	Local Control Panel
LCV	Level Control Valve
LP	Low Pressure
MSDS	Material Safety Data Sheet
MTPA	Metric Tonnes per Annum
MTPD	Metric Tonnes Per Day
NAG	Non-Associated Gas
NUPRC	Nigerian Upstream Petroleum Regulatory Commission
OML	Operating Mining License

1. Introduction

1.1. Background Information

Brass Fertilizer and Petrochemical Company Limited (BFPCL) is developing a 10,000 metric tonnes per day (MTPD) capacity greenfield methanol production facility in Akabel (Odioma), Brass LGA, Bayelsa state. The plant complex is to be based on natural gas feedstock from 5 SPDC fields in OMLs 33, 35 & 36.

The Project involves the development, construction, and operation of a plant comprising:

- Two trains of 5,000 MTPD methanol plant, producing a total of 10,000 MTPD of methanol
- 340 MMscfd Gas Processing Plant ("GPP"), which will extract NGL from the natural gas feedstock, prior to feeding the lean gas to the methanol plant.
- Gas gathering pipelines and associated manifolds transporting the natural gas from five fields, viz: Bubouwe Bou (BBOU), Elepa (ELEP), Nun River (NUNR), Seibou (SEIB), and Opugbene (OPUG) to the GPP at Akabel.
- Product Export Lines and SBM for the export of the methanol, and NGL of the GPP.
- All offsite, utility, and off-plot facilities (including product storage, internal power generation, steam generation, and export jetty) for the operation of the processing plants.

The Pipelines will be implemented in two phases, - starting with BBOU & ELEP as Phase 1 and followed by NUNR, SEIB & OPUG as Phase 2.

The focus of the present FEED is on Phase 1 as shown in figure 1 below, while Phase 2 will be implemented later.

TCE is the Project Management Consultant for the project.

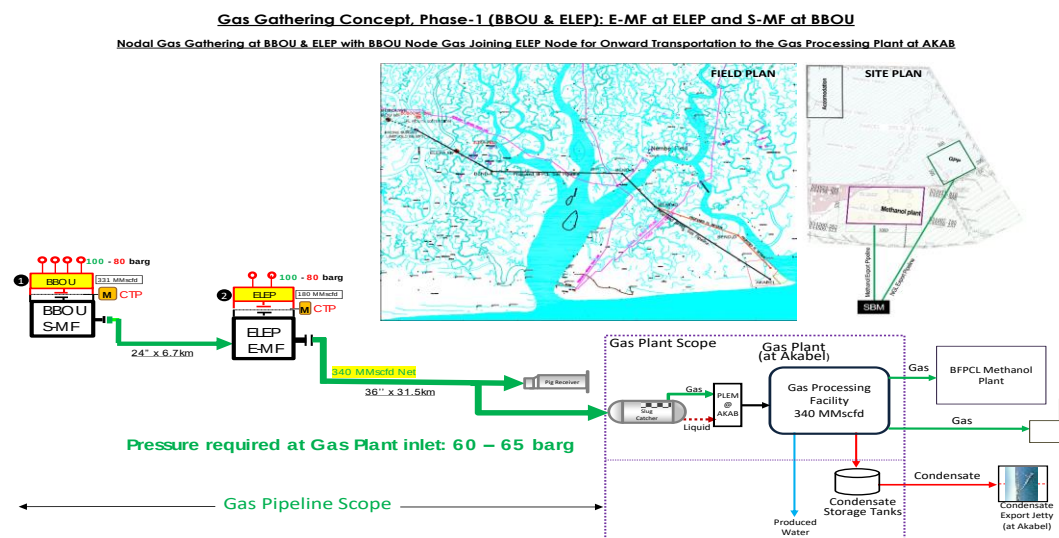


Figure 1: Gas Gathering Concept, Phase 1

1.2. Purpose

The purpose of this document is to define the consumables requirements and the supply plan for the operation of the gas gathering pipelines.

The FEED shall fully comply with all specified relevant contractual requirements.

2. Regulations, Codes And Standards

The latest versions of the regulations, codes, and standards and extant amendments or supplements shall apply. Local laws regulating the oil and gas industry in Nigeria shall be strictly adhered to. The order of precedence for the documents shall be as follows:

- Nigerian Local Standards and Codes
- The Contract
- Project Specifications
- Employer's Standards
- Shell DEPs
- International Codes and Standards

In all cases, the most stringent shall apply except where constraints dictate otherwise.

Conflicts between specifications and referenced Codes and Standards or deviations thereof shall be referred in a Technical Query (TQ) to the EMPLOYER for resolution.

3. Consumables Requirements

This consumables requirements and supply plan identifies the consumables that would be required for routine operations of the Pipelines and Associated Infrastructure and specifies the plan for the sourcing, supply to site and filling of the relevant site storage tanks. The plan is based on the need to ensure availability of the consumable when required.

This document shall be updated during the detailed engineering phase of the project when the design and operating requirements of the facilities are fully detailed.

3.1. List of Consumables

The only identified consumable at this stage is Corrosion Inhibitor. The type will be selected during detailed engineering design based on certain criteria like historical performance, availability locally, cost and ease of handling.

There is no gas processing or conditioning, or any form of phase separation that will take place within the Pipelines and Associated Infrastructure. In addition, the FEED has been based on a philosophy that simplifies the process and reduces or eliminates complex equipment and accessories in the field manifolds because of their remoteness and operating philosophy of being unmanned. As a result of this deliberate effort to simplify the facilities, other commonly used consumables in the oil and gas industry are not envisaged for the Pipelines and Associated Infrastructure. These consumables include.

- lube oil (for rotating equipment),
- diesel (for diesel engines),
- de-emulsifier,
- anti-foam,
- glycol (MEG, DEG or TEG for gas dehydration),
- hydrate inhibitor, and
- biocide

4. Supply Plan

4.1. Sourcing of Consumables

Delivery schedule will be a key factor in selecting a supplier, or supply source, for consumables. In this regard, local suppliers will be considered first before foreign suppliers.

Using local suppliers will also fulfill a critical requirement of the Nigeria Content Development Act which aims to domesticate work in the oil and gas industry as much as possible

Given the criticality of consumables to the operation of the facilities, it is advisable to enter into medium-term and even long-term supply agreements with the suppliers with well-defined key performance indicators to guarantee availability of the product as and when required.

Re-order levels will be established for the consumables based on rate of consumption, delivery schedule, ease of handling, safe storage capacity and shelf life.

4.2. Delivery to Site

It is envisaged that there will be a warehouse within the Methanol Complex at Akabel where significant quantities of consumables, say up to 6 months requirement, can be stored. The warehouse should be equipped with the necessary accessories to meet the storage conditions (humidity, temperature, ventilation, etc.) stipulated by the manufacturers.

The corrosion Inhibitor, like most chemicals, will be stored in plastic drums and transported from Akabel to the field manifolds at BBOU and ELEP by boat or barge.

4.3. Offloading at Site

At each field manifold, a jib crane will be provided at the boat landing to lift heavy items from boats and barges. The jib crane will be adequately sized to safely lift the heaviest item that will be expected to be delivered by boat or barge for normal operations.

The corrosion inhibitor drums could either be rolled on the floor from the boat landing, or pushed on a trolley, to the corrosion inhibitor skid where the storage tank will be located. The storage tank will be fitted with a quick coupling connection for connection of a hose for filling.

Once at the skid area, the corrosion inhibitor will be transferred from the drums to the storage tank by means of a portable, or fixed, electric pump and a hose fitted with a matching quick coupling connection to the one on the tank. The storage tank will be sized to store at least one month's (30 days') supply of corrosion inhibitor.

It could also be designed such that transfer of corrosion inhibitor from the drum to the storage tank can take place straight from the boat landing by means of a portable electric pump and long hose.

The handling of the corrosion inhibitor must be in strict compliance with the Material Safety Data Sheet (MSDS).